

Advanced (Habanero)

- Q1.** A bakery sells x loaves of bread and y pastries. The profit is $3x + 2y$. If the profit is at least

$$180$$

, which inequality represents this situation?

- (A) $3x + 2y \geq 180$
 (B) $3x - 2y \geq 180$
 (C) $2x + 3y \geq 180$
 (D) $2x - 3y \geq 180$
 (E) $x + y \geq 180$

- Q2.** A company produces two products, x and y , with costs

$$5x +$$

$3y$ and $2x + 4y$ respectively. If the total cost is at most 1500, which inequality represents this situation?

- (A) $5x + 3y \leq 1500$
 (B) $5x - 3y \leq 1500$
 (C) $2x + 4y \leq 1500$
 (D) $2x - 4y \leq 1500$
 (E) $x + y \leq 1500$

- Q3.** A farmer has x acres of wheat and y acres of corn. The total area is 100 acres. Which inequality represents the possible values of x and y ?

- (A) $x + y \leq 100$
 (B) $x + y \geq 100$
 (C) $x - y \leq 100$
 (D) $x - y \geq 100$
 (E) $x \cdot y \leq 100$

- Q4.** A city has a maximum of 5000 people in a concert. If x is the number of adults and y is the number of children, which inequality represents this situation?

- (A) $x + y \leq 5000$
 (B) $x + y \geq 5000$
 (C) $x - y \leq 5000$
 (D) $x - y \geq 5000$
 (E) $x \cdot y \leq 5000$

- Q5.** A company produces two products, x and y . The profit from x is

$$10$$

and from y is

$$15$$

. If the total profit is at least

$$300$$

, which inequality represents this situation?

- (A) $10x + 15y \geq 300$
 (B) $10x - 15y \geq 300$
 (C) $15x + 10y \geq 300$
 (D) $15x - 10y \geq 300$
 (E) $x + y \geq 300$

- Q6.** A student has at most

$$20$$

to spend on pens and pencils. If x is the number of pens and y is the number of pencils, and each pen costs

$$2$$

and each pencil costs

$$1$$

, which inequality represents this situation?

- (A) $2x + y \leq 20$
 (B) $2x - y \leq 20$
 (C) $x + 2y \leq 20$
 (D) $x - 2y \leq 20$
 (E) $x \cdot y \leq 20$

- Q7.** A factory produces x units of product A and y units of product B . The cost of producing A is

$$5$$

and B is

$$3$$

. If the total cost is at most

$$150$$

, which inequality represents this situation?

- (A) $5x + 3y \leq 150$
 (B) $5x - 3y \leq 150$
 (C) $3x + 5y \leq 150$
 (D) $3x - 5y \leq 150$
 (E) $x + y \leq 150$

- Q8.** A store has a maximum of 200 items to display. If x is the number of shirts and y is the number of pants, which inequality represents this situation?

- (A) $x + y \leq 200$
 (B) $x + y \geq 200$
 (C) $x - y \leq 200$
 (D) $x - y \geq 200$
 (E) $x \cdot y \leq 200$

Open Ended Questions (FRQ)

- Q9.** A company produces two products, x and y , with costs

$$4x +$$

$2y$ and $3x + 5y$ respectively. If the total cost is at most 1000 and the profit from x is

$$10$$

and from y is

$$15$$

, find the feasible region and optimize the profit within this region.

Q10. A farmer has x acres of wheat and y acres of corn. The total area is 150 acres. If the profit from wheat is 50 and from corn is 75, and the cost of producing wheat is 10 and corn is 20, find the feasible region and optimize the profit within this region.

Chef's Tips

- Tip 1: Read the questions carefully
- Tip 2: Use the given information to find the feasible region
- Tip 3: Optimize the profit within the feasible region